week1 Utility

- (a)A person has $20 and a logarithmic utility function $u(w)=\ln w$. He is considering buying a ticket to enter a game. He wins with probability 50% and loses with probability 50%. If he wins, he receives a prize of $20; if he loses, he receives nothing. What is the highest price he is willing to pay for the ticket?

  Let p denote the maximum price he is willing to pay. After buying the ticket, final wealth is $w_1=40-p$ if he wins and $w_0=20-p$ if he loses. Indifference between buying and not buying requires

  $$\frac{1}{2}\ln(40-p)+\frac{1}{2}\ln(20-p)=\ln 20.$$

  Combining the logs,

  $$\ln\left(\sqrt{(40-p)(20-p)}\right)=\ln 20,$$

  hence

  $$(40-p)(20-p)=400.$$

  This simplifies to

  $$p^2-60p+400=0,$$

  whose roots are

  $$p=30\pm 10\sqrt{5}.$$

  Because a feasible ticket price must satisfy $p\leq 20$, the answer is

  $$p^*=30-10\sqrt{5}\approx 7.64.$$

  (b)Let $\tilde{y}=e^{\tilde{x}}$ where $\tilde{x}\sim N(\mu,\sigma^2)$. Show that

  $$\frac{\operatorname{stdev}(\tilde{y})}{\mathbb{E}[\tilde{y}]}=\sqrt{e^{\sigma^2}-1}.$$

  Since $\tilde{x}\sim N(\mu,\sigma^2)$, its moment generating function is

  $$M_{\tilde{x}}(t)=\mathbb{E}[e^{t\tilde{x}}]=\exp\left(\mu t+\tfrac{1}{2}\sigma^2t^2\right).$$

  Setting $t=1$ and $t=2$ gives

  $$\mathbb{E}[\tilde{y}]=\mathbb{E}[e^{\tilde{x}}]=e^{\mu+\frac{\sigma^2}{2}},\qquad \mathbb{E}[\tilde{y}^{\,2}]=\mathbb{E}[e^{2\tilde{x}}]=e^{2\mu+2\sigma^2}.$$

  Hence

  $$\operatorname{Var}(\tilde{y})=\mathbb{E}[\tilde{y}^{\,2}]-(\mathbb{E}[\tilde{y}])^2=e^{2\mu+2\sigma^2}-e^{2\mu+\sigma^2}=e^{2\mu+\sigma^2}\left(e^{\sigma^2}-1\right).$$

  Therefore,

  $$\operatorname{stdev}(\tilde{y})=\sqrt{\operatorname{Var}(\tilde{y})}=e^{\mu+\frac{\sigma^2}{2}}\sqrt{e^{\sigma^2}-1}.$$

  Dividing by $\mathbb{E}[\tilde{y}]=e^{\mu+\frac{\sigma^2}{2}}$ yields

  $$\frac{\operatorname{stdev}(\tilde{y})}{\mathbb{E}[\tilde{y}]}=\sqrt{e^{\sigma^2}-1}.$$

week2 SDF

- There are two states of the world, ω_1 and ω_2. There are two assets:
  (1) a risk-free asset that pays R_f in each state; its date-0 price is normalized to 1 and its date-1 payoff per share is R_f;
  (2) a risky asset with date-0 price 1 and date-1 payoff x. Let $R_d=x(\omega_1)$ and $R_u=x(\omega_2)$ with $R_u>R_d$.
  Answer:

(a) What inequalities among R_f, R_d, R_u are equivalent to no-arbitrage?
(b) Assuming no arbitrage, compute the unique risk-neutral probabilities of ω_1 and ω_2.
(c) Suppose a new asset is introduced that pays $\max(x-K,0)$ for a constant K. Find its date-0 price under no-arbitrage.

(a) No-arbitrage inequalities

If $R_f \le R_d < R_u$, go long the risky asset and short the risk-free asset. The net payoff is

$$x - R_f \ge R_d - R_f \ge 0,$$

strictly positive in some state, yielding arbitrage.

If $R_f \ge R_u > R_d$, short the risky asset and go long the risk-free asset. The net payoff is

$$R_f - x \ge R_f - R_u \ge 0,$$

again an arbitrage.

Hence no-arbitrage requires R_f to lie strictly between the two state payoffs:

$$\boxed{R_u > R_f > R_d}.$$

(b) Risk-neutral probabilities

Let $q = \mathbb{Q}(\omega_1)$ and $1-q = \mathbb{Q}(\omega_2)$. Pricing the risky asset gives

$$1 = \frac{1}{R_f}\big(qR_d + (1-q)R_u\big),$$

so

$$q = \frac{R_u - R_f}{R_u - R_d}, \qquad 1-q = \frac{R_f - R_d}{R_u - R_d}.$$

(c) Price of the option

For $c_1 = \max(x-K,0)$, the date-0 price is

$$C_0 = \frac{1}{R_f}\Big(q\max(R_d-K,0) + (1-q)\max(R_u-K,0)\Big).$$

By cases:

If $K \le R_d < R_u$,

$$C_0 = \frac{1}{R_f}\left(\frac{R_u-R_f}{R_u-R_d}(R_d-K) + \frac{R_f-R_d}{R_u-R_d}(R_u-K)\right).$$

If $R_d < K < R_u$,

$$C_0 = \frac{1}{R_f}\frac{R_f-R_d}{R_u-R_d}(R_u-K).$$

If $R_d < R_u \le K$, then $C_0 = 0$.

week3 mean-variance analysis

- (a)

  Suppose there are two risky assets with expected returns

  $$\mu_1 = 1.08, \quad \mu_2 = 1.16,$$

  standard deviations

  $$\sigma_1 = 0.25, \quad \sigma_2 = 0.35,$$

  and correlation

  $$\rho_{12} = 0.30.$$

Construct the global minimum-variance (GMV) portfolio using these two assets. Question: What are the expected return and standard deviation of this portfolio?

Given $\mu_1 = 1.08,\ \mu_2 = 1.16,\ \sigma_1 = 0.25,\ \sigma_2 = 0.35,\ \rho_{12} = 0.30$,
the covariance is $\sigma_{12} = \rho_{12}\sigma_1\sigma_2 = 0.02625$.
Let $w_2 = 1 - w_1$. The variance is

$$\sigma_p^2 = w_1^2\sigma_1^2 + (1-w_1)^2\sigma_2^2 + 2w_1(1-w_1)\sigma_{12}.$$

FOC yields the closed form:

$$w_1^* = \frac{\sigma_2^2 - \sigma_{12}}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}}, \quad w_2^* = 1 - w_1^* = \frac{\sigma_1^2 - \sigma_{12}}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}}.$$

Plugging $\sigma_1^2 = 0.0625,\ \sigma_2^2 = 0.1225$ gives

$$(w_1^*, w_2^*) = (0.726415,\ 0.273585), \quad \sigma_{\text{GMV}}^2 = 0.0525825\ (\sigma_{\text{GMV}} = 0.229309), \quad \mu_{\text{GMV}} = 1.101887.$$

Based on (a), now consider a third risky asset with expected return

$$\mu_3 = 1.10,$$

and standard deviation

$$\sigma_3 = 0.30,$$

which is independent of the first two assets (zero covariances).Using these three risky assets, derive the efficient frontier.

Add asset 3 with $\mu_3 = 1.10,\ \sigma_3 = 0.30$, independent of the first two.

$$\Sigma = \begin{bmatrix} 0.0625 & 0.02625 & 0 \\ 0.02625 & 0.1225 & 0 \\ 0 & 0 & 0.09 \end{bmatrix}, \quad \mu = \begin{bmatrix} 1.08 \\ 1.16 \\ 1.10 \end{bmatrix}, \quad \mathbf{1} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

Define

$$A = \mathbf{1}^\top\Sigma^{-1}\mathbf{1} = 30.128828,\ B = \mathbf{1}^\top\Sigma^{-1}\mu = 33.177593,\ C = \mu^\top\Sigma^{-1}\mu = 36.583193,\ D = AC - B^2 = 1.456031.$$

Then the GMV portfolio is

$$w_{\text{GMV}} = \frac{\Sigma^{-1}\mathbf{1}}{A} = \begin{bmatrix} 0.458523 \\ 0.172690 \\ 0.368787 \end{bmatrix}, \quad \mu_{\text{GMV}} = \frac{B}{A} = 1.101191, \quad \sigma_{\text{GMV}}^2 = \frac{1}{A} = 0.0331908.$$

The (minimum) variance for a target return μ_p is

$$\sigma_p^2(\mu_p) = \frac{A\mu_p^2 - 2B\mu_p + C}{D},$$

and the optimizer is

$$w(\mu_p) = \frac{C - B\mu_p}{D}\,\Sigma^{-1}\mathbf{1} + \frac{A\mu_p - B}{D}\,\Sigma^{-1}\mu.$$

Example: for $\mu_p = 1.12$,
$w = (0.227273, 0.409091, 0.363636)$ and $\sigma_p = 0.201274$.

week4 multiple factor model

- (a)Assume the return of the tangency portfolio on the mean-variance frontier, $\tilde{R}_{\text{tang}}$, is an affine function of a vector $\tilde{F}$; that is,

$$\tilde{R}_{\text{tang}} = a + b'\tilde{F}.$$

Assume there is a risk-free asset with the return R_f. Show that there is a factor model with factors $\tilde{F}$.

Assume a risk-free asset with return R_f exists and the tangency portfolio return satisfies

$$\widetilde{R}_{\text{tang}} = a + b'\widetilde{F}.$$

With a risk-free asset, any asset i admits the tangency/CAPM projection

$$\widetilde{R}_i = R_f + \beta_i(\widetilde{R}_{\text{tang}} - R_f) + \tilde{\varepsilon}_i, \quad \beta_i = \frac{\text{Cov}(\widetilde{R}_i, \widetilde{R}_{\text{tang}})}{\text{Var}(\widetilde{R}_{\text{tang}})},$$

with $\mathbb{E}[\tilde{\varepsilon}_i] = 0$ and $\text{Cov}(\tilde{\varepsilon}_i, \widetilde{R}_{\text{tang}}) = 0$.
Substituting $\widetilde{R}_{\text{tang}} = a + b'\widetilde{F}$ yields

$$\widetilde{R}_i = \underbrace{(R_f + \beta_i(a - R_f))}_{\alpha_i} + \underbrace{(\beta_i b)'}_{\gamma_i}\widetilde{F} + \tilde{\varepsilon}_i,$$

which is a K-factor (statistical) model:

$$\widetilde{R}_i = \alpha_i + \gamma_i'\widetilde{F} + \tilde{\varepsilon}_i.$$

If desired, one may orthogonalize residuals by projecting $\widetilde{R}_i$ on $\{1, \widetilde{F}\}$ to obtain a residual uncorrelated with factors.

(b)Suppose two well-diversified risky portfolios satisfy a statistical factor model with a single factor:

$$\tilde{R}_1 = 1.02 + \tilde{f}, \qquad \tilde{R}_2 = \mu_2 - \tilde{f},$$

where μ_2 is a constant parameter. Further, there is a risk-free asset with return of $R_f = 1.04$.

(i)If $\mu_2 = 1.05$, is there any arbitrage opportunity? If yes, then how do you arbitrage?

(ii)If $\mu_2 = 1.08$, is there any arbitrage opportunity? If yes, then how do you arbitrage?

Given

$$\widetilde{R}_1 = 1.02 + \tilde{f}, \qquad \widetilde{R}_2 = \mu_2 - \tilde{f}, \qquad R_f = 1.04,$$

the equal-weight portfolio

$$\widetilde{R}_p = \tfrac{1}{2}\widetilde{R}_1 + \tfrac{1}{2}\widetilde{R}_2 = \tfrac{1}{2}(1.02 + \mu_2)$$

is riskless (the factor cancels pathwise).

(i) $\mu_2 = 1.05$
Then $\widetilde{R}_p = 1.035 < 1.04$. Arbitrage: short the equal-weight synthetic riskless portfolio and lend at R_f; payoff $1.04 - 1.035 = 0.005 > 0$ per \$1 notional.

(ii) $\mu_2 = 1.08$.

Then $\widetilde{R}_p = 1.05 > 1.04$. Arbitrage: go long the equal-weight synthetic riskless portfolio and borrow at R_f; payoff $1.05 - 1.04 = 0.01 > 0$ per \$1 notional.

week5 information in finance

- In the Grossman--Stiglitz economy, assume the uninformed investors are risk neutral. Find a partially revealing equilibrium in which the price reveals

$$\tilde{\omega} = \tilde{s} - \delta\tilde{z}.$$

Solution:

Let $\tilde{s} = \tilde{w} + \tilde{\varepsilon}$, where $\tilde{w}$ is the asset's true fundamental value,
$\tilde{s}$ is the informed investors' private signal, and $\tilde{\varepsilon}$ is market noise.

Construct

$$\tilde{r} = \tilde{s} - \delta\tilde{z},$$

where $\tilde{r}$ is a ``purified'' signal that mixes the private signal $\tilde{s}$ with the noise $\tilde{z}$. The price p does not reveal $\tilde{w}$ directly; instead, it reveals $\tilde{r}$.

If p were in a one-to-one correspondence with the true value $\tilde{w}$, no one would pay for information and the equilibrium would not exist. Hence the equilibrium pricing rule is

$$p = \mu_w + k(\tilde{r} - \mu_w),$$

where p is the equilibrium price, $\mu_w = \mathbb{E}[\tilde{w}]$ is the prior mean of the fundamental, and $k \in (0,1)$ measures the sensitivity of price to the revealed statistic $\tilde{r}$.

Substituting $\tilde{r} = \tilde{s} - \delta\tilde{z}$ gives

$$p = \mu_w + k(\tilde{s} - \delta\tilde{z} - \mu_w).$$

Here $k \neq 0$ means the price p carries information; $k < 1$ means the price moves with $\tilde{r}$ but not one-for-one because of noise.

Moreover,

$$p = \mathbb{E}[\tilde{w} \mid \tilde{p}] = \mathbb{E}[\tilde{w} \mid \tilde{r}],$$

since p and $\tilde{r}$ are in one-to-one correspondence and $(\tilde{w}, \tilde{r})$ are jointly normal. Therefore (the standard linear-projection formula),

$$\mathbb{E}[\tilde{w} \mid \tilde{r}] = \mu_w + \frac{\operatorname{Cov}(\tilde{w}, \tilde{r})}{\operatorname{Var}(\tilde{r})}\big(\tilde{r} - \mathbb{E}[\tilde{r}]\big).$$

Because

$$\tilde{r} = \tilde{s} - \delta\tilde{z} = \tilde{w} + \tilde{\varepsilon} - \delta\tilde{z},$$

and $\tilde{\varepsilon}, \tilde{z}$ are independent of $\tilde{w}$, we have

$$\operatorname{Cov}(\tilde{w}, \tilde{r}) = \operatorname{Cov}(\tilde{w}, \tilde{w} + \tilde{\varepsilon} - \delta\tilde{z}) = \sigma_w^2, \qquad \operatorname{Var}(\tilde{r}) = \operatorname{Var}(\tilde{w} + \tilde{\varepsilon} - \delta\tilde{z}) = \sigma_w^2 + \sigma_\varepsilon^2 + \delta^2\sigma_z^2,$$

$$\mathbb{E}[\tilde{r}] = \mathbb{E}[\tilde{w}] + \mathbb{E}[\tilde{\varepsilon}] - \delta\,\mathbb{E}[\tilde{z}] = \mu_w \quad (\text{both } \tilde{\varepsilon} \text{ and } \tilde{z} \text{ are zero-mean noise}).$$

Plugging back yields

$$p = \mu_w + \frac{\sigma_w^2}{\sigma_w^2 + \sigma_\varepsilon^2 + \delta^2\sigma_z^2}\,(\tilde{r} - \mu_w).$$

Thus the price fluctuates around the true value but does not equal it; the amplitude is governed by the weight. The weight increases with information quality (smaller σ_w^2 implies higher precision and a larger weight) and decreases with noise intensity (larger $\delta^2\sigma_z^2$ reduces the weight).

Since

$$k = \frac{\sigma_w^2}{\sigma_w^2 + \sigma_\varepsilon^2 + \delta^2\sigma_z^2},$$

as long as

$$\sigma_\varepsilon^2 + \delta^2\sigma_z^2 > 0,$$

we have $k \in (0,1)$: price reflects only part of the information and is not perfectly revealing.

Finally, from

$$p = \mu_w + k(\tilde{r} - \mu_w),$$

we can invert to obtain

$$\tilde{r} = \mu_w + \frac{p - \mu_w}{k}.$$

When $k < 1$, the price underreacts to information (attenuation); recovering $\tilde{r}$ from p requires amplification by a factor of $1/k$.

- In the Kyle model, assume the informed investor has CARA utility.
  There is a linear equilibrium. Derive an expression for λ as a root of a fifth-order polynomial.

Solution:

Setups:

Let the fundamental value satisfy $\tilde{v} \sim N(\mu, \Sigma_v)$ and noise orders $\tilde{u} \sim N(0, \Sigma_u)$. The informed investor observes $\tilde{v}$ and chooses an order x.
The total order flow is $y = x + \tilde{u}$. A competitive, risk–neutral market maker sets a linear price

$$p = \mu + \lambda y.$$

The informed investor has CARA utility with absolute risk–aversion $\varphi > 0$.
A CARA utility takes the form

$$U(W) = -e^{-\varphi W}, \qquad \varphi > 0,$$

and its absolute risk aversion is

$$A(W) = -\frac{U''(W)}{U'(W)} = \varphi,$$

a constant.

\section*{Mean--variance equivalence under normality}
We wish to choose x to maximize

$$\max\ \mathbb{E}[U(W)],$$

which is equivalent to maximizing

$$\mathbb{E}[W \mid \tilde{v}] - \frac{\varphi}{2}\mathrm{Var}[W \mid \tilde{v}],$$

where $W = x(\tilde{v} - p)$.

\textit{Proof sketch.}
If $W \sim N(\mu, \sigma^2)$, then

$$\mathbb{E}[U(W)] = \mathbb{E}\left[-e^{-\varphi W}\right] = -M_W(-\varphi),$$

where $M_W(t) = \mathbb{E}[e^{tW}]$ is the moment generating function. For a normal random variable,

$$M_W(t) = \exp\left(\mu t + \tfrac{1}{2}\sigma^2 t^2\right),$$

so

$$\mathbb{E}[U(W)] = -\exp\left(-\varphi\mu + \tfrac{1}{2}\varphi^2\sigma^2\right).$$

Because the exponential is monotone, maximizing $\mathbb{E}[U(W)]$ is equivalent to maximizing $-\varphi\mu + \frac{1}{2}\varphi^2\sigma^2$, i.e., maximizing $\mu - \frac{1}{2}\varphi\sigma^2$.

\section*{Informed trader's objective}
Substituting $p = \mu + \lambda(x + \tilde{u})$ gives per–share profit

$$W = x\left(\tilde{v} - \mu - \lambda x - \lambda\tilde{u}\right).$$

Hence

$$\begin{aligned}
\mathbb{E}[W \mid \tilde{v}] &= \mathbb{E}[x(\tilde{v} - \mu - \lambda x - \lambda\tilde{u}) \mid \tilde{v}] \\
&= x(\tilde{v} - \mu - \lambda x) - x\lambda\mathbb{E}[\tilde{u}] \\
&= x(\tilde{v} - \mu - \lambda x),
\end{aligned}$$

$$\begin{aligned}
\mathrm{Var}[W \mid \tilde{v}] &= \mathrm{Var}[x(\tilde{v} - \mu - \lambda x - \lambda\tilde{u}) \mid \tilde{v}] \\
&= \mathrm{Var}[-x\lambda\tilde{u} \mid \tilde{v}] = \lambda^2 x^2\,\mathrm{Var}[\tilde{u}] = \lambda^2 x^2 \Sigma_u.
\end{aligned}$$

The first–order condition is therefore

$$(\tilde{v} - \mu) - 2\lambda x - \varphi\lambda^2\Sigma_u\, x = 0,$$

which yields

$$x(\tilde{v}) = \frac{\tilde{v} - \mu}{2\lambda + \varphi\lambda^2\Sigma_u}.$$

Thus the optimal position x is proportional to the signal deviation $\tilde{v} - \mu$.
Define

$$\beta = \frac{1}{2\lambda + \varphi\lambda^2\Sigma_u}, \qquad x = \beta(\tilde{v} - \mu).$$

\section*{Market maker's rational expectations}
The order flow is

$$y = x + \tilde{u} = \beta(\tilde{v} - \mu) + \tilde{u}.$$

Since

$$p = \mathbb{E}[\tilde{v} \mid y] = \mathbb{E}[\tilde{v}] + \frac{\mathrm{Cov}(\tilde{v}, y)}{\mathrm{Var}(y)}\bigl(y - \mathbb{E}[y]\bigr) \quad \text{and} \quad p = \mu + \lambda y,$$

we must have

$$\lambda = \frac{\mathrm{Cov}(\tilde{v}, y)}{\mathrm{Var}(y)}.$$

Compute

$$\begin{aligned} \mathrm{Cov}(\tilde{v}, y) &= \mathrm{Cov}\left(\tilde{v}, \beta(\tilde{v} - \mu) + \tilde{u}\right) = \beta\,\mathrm{Cov}(\tilde{v}, \tilde{v} - \mu) = \beta\Sigma_v, \\ \mathrm{Var}(y) &= \mathrm{Var}\bigl[\beta(\tilde{v} - \mu) + \tilde{u}\bigr] = \beta^2\Sigma_v + \Sigma_u. \end{aligned}$$

Hence

$$\lambda = \frac{\beta\Sigma_v}{\beta^2\Sigma_v + \Sigma_u}.$$

Substituting β and letting $D = 2\lambda + \varphi\lambda^2\Sigma_u$, we have

$$\lambda = \frac{(\Sigma_v/D)}{(\Sigma_v/D^2) + \Sigma_u} = \frac{\Sigma_v D}{\Sigma_v + \Sigma_u D^2}.$$

\section*{Fifth--order polynomial for λ}
Expanding $D^2 = (2\lambda + \varphi\lambda^2\Sigma_u)^2$ and rearranging yields the quintic

$$\varphi^2\Sigma_u^3\,\lambda^5 + 4\varphi\Sigma_u^2\,\lambda^4 + 4\Sigma_u\,\lambda^3 - \varphi\Sigma_u\Sigma_v\,\lambda^2 - \Sigma_v\,\lambda = 0.$$

Discarding the trivial solution $\lambda = 0$ (no price impact), the economically relevant solution is the unique positive root. In the limit $\varphi \to 0$ (the informed trader becomes risk–neutral), the equation reduces to

$$4\Sigma_u\,\lambda^3 - \Sigma_v\,\lambda = 0,$$

whose nonzero solution coincides with the classic Kyle slope $\lambda = \frac{1}{2}\sqrt{\Sigma_v/\Sigma_u}$.